UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

THIRD-YEAR FINAL EXAMINATION

03 01129

LH STATISTICAL PHYSICS

SUMMER EXAMINATIONS 2025

Time allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

A formula sheet and a table of physical constants and units that may be required will be found at the end of this question paper.

No calculator is permitted in this examination.

Answer **two** *questions. If you answer more than two questions, credit will only be given for the best two answers.*

1. Under certain conditions, the surface waves in liquid helium are quantised and may be considered as quasi-particles with a dispersion law $\varepsilon(\mathbf{p}) = (|\mathbf{p}|/\alpha)^{3/2}$ where $\mathbf{p}$ is the momentum of the quasi-particle and α is a constant. Such quasi-particles are called ripplons. **For the purposes of the question, they may be treated as non-degenerate, i.e. classical, bosons.**

   Consider a **two**-dimensional ideal gas of N ripplons in thermal equilibrium at temperature T confined to an area A.

   (a) Show that the one-particle density of states, $n(\varepsilon)$, of the ripplon gas is

   $$n(\varepsilon) = \frac{A}{\mathrm{h}^2}\frac{4\pi}{3}\alpha^2\varepsilon^{1/3} .$$

   [8]

   (b) Show that the one-particle partition function for the ripplon gas

   $$Z_1 = \frac{4\pi\Gamma(4/3)}{3}\frac{\alpha^2 A}{\mathrm{h}^2}\beta^{-(4/3)} ,$$

   where the gamma function, $\Gamma(x)$, is defined on the formula sheet. Assuming that ripplons are indistinguishable, explain why and under what conditions the total partition function Z is given by

   $$Z = \frac{Z_1^N}{N!} .$$

   [9]

   (c) Use the partition function above to calculate the free energy F and thus the entropy S of the gas. **[8]**

   (d) The entropy S found above becomes negative for $n = N/V > n_0$, where n_0 is a certain critical density. Find, neglecting numerical factors, the temperature dependence of n_0. Explain which of the assumptions made in the model became invalid for large densities. **[5]**

2. A monatomic gas is in equilibrium with the surface of a solid. The surface is a two-dimensional lattice of N_0 sites which can be unoccupied, or occupied by one of the gas atoms. The partition function of an unoccupied site is 1, and of an occupied site is $z(T)$. The atoms adsorbed onto the surface do not interact with each other.

   (a) If N atoms are adsorbed onto the surface, explain why the partition function is given by
   $$Z_{\mathrm{s}}(N,T) = \frac{N_0!}{N!(N_0-N)!}\left[z(T)\right]^N \ .$$ **[6]**

   (b) Derive an expression for the Helmholtz free energy, $F_{\mathrm{s}}(N,T)$, and use it to show that the chemical potential of the adsorbed atoms is
   $$\mu_{\mathrm{s}}(N,T) = \mathrm{k_B}T\ln\left[\frac{N}{(N_0-N)z(T)}\right] .$$ **[6]**

   The partition function of an ideal gas of M monatomic atoms of mass m in a volume V is given by
   $$Z_{\mathrm{g}}(M,T,V) = \frac{V^M}{M!\,\lambda_{\mathrm{T}}^{3M}} \qquad \text{where} \qquad \lambda_{\mathrm{T}} = \frac{\mathrm{h}}{\sqrt{2\pi m\mathrm{k_B}T}} \ .$$

   (c) Write down an expression for the Helmholtz free energy of the gas, $F_{\mathrm{g}}(M,T,V)$, and use it to derive the equation of state of the gas, $pV = M\mathrm{k_B}T$. **[5]**

   (d) Using the results derived in part (c), show that the chemical potential for the gas is
   $$\mu_{\mathrm{g}}(M,T,V) = \mathrm{k_B}T\ln\left[\frac{M\lambda_{\mathrm{T}}^3}{V}\right] .$$ **[5]**

   (e) By considering the equilibrium between the gas atoms and those adsorbed on the surface, show that the coverage of the surface, $\theta = N/N_0$, is related to the gas pressure by
   $$\theta = \frac{N}{N_0} = \frac{p}{p+p_0(T)} \ ,$$
   and derive an expression for $p_0(T)$. **[8]**

3. In this question we consider a **two**-dimensional gas of N non-interacting, non-relativistic, electrons of mass m in an area A, which are in thermal equilibrium at temperature T. The following mathematical results will be needed:

$$\int \mathrm{d}x\ \left(\mathrm{e}^{x}+1\right)^{-1} = -\ln\left(1+\mathrm{e}^{-x}\right)\ ,$$

$$\int_0^\infty \mathrm{d}x\ \ln\left(1+\mathrm{e}^{-x}\right) = \int_0^\infty \mathrm{d}x\ \sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\mathrm{e}^{-nx} = \sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}\ .$$

(a) Show that the one-particle density of states, $n(\varepsilon)$, for the electron gas is

$$n(\varepsilon) = \begin{cases} \dfrac{mA}{\pi\hbar^2} & \text{for } \varepsilon > 0 \\ 0 & \text{for } \varepsilon < 0\ . \end{cases}$$

[7]

(b) Use the result of part (a) to show that the Fermi energy, ε_{F}, of the electron gas is

$$\varepsilon_{\mathsf{F}} = \frac{\pi\hbar^2 n}{m}\ ,$$

where $n = N/A$ is the areal density of electrons. **[5]**

(c) Show that the chemical potential at finite temperature, $\mu(T)$, is given by

$$\mathrm{e}^{\beta\mu} = \mathrm{e}^{\beta\varepsilon_{\mathsf{F}}} - 1\ .$$

Find the first two terms in the expression, in powers of T, for $\mu(T)$ at low temperatures. **[8]**

(d) Write down an expression for the internal energy U in terms of $n(\varepsilon)$ and the Fermi function $f(\varepsilon)$. Use integration by parts to show that

$$U = -\Omega = \mathrm{k_B}T\int_0^\infty \mathrm{d}\varepsilon\ n(\varepsilon)\ln\left(1+\mathrm{e}^{-\beta(\varepsilon-\mu)}\right)\ ,$$

where Ω is the grand potential. **[5]**

(e) Use the result of part (d) to show that

$$U = N\frac{\mu^2}{2\varepsilon_{\mathsf{F}}} + N\frac{(\mathrm{k_B}T)^2}{\varepsilon_{\mathsf{F}}}\left\{\frac{\pi^2}{12} + \sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\mathrm{e}^{-n\beta\mu}\right\}\ .$$

[5]

Formula Sheet

The first law of thermodynamics

$$\mathrm{d}U = T\mathrm{d}S - p\,\mathrm{d}V$$

Thermodynamic functions of state

$F = U - TS$	(Helmholtz) Free Energy
$H = U + pV$	Enthalpy
$G = F + pV$	Gibbs Free Energy

The first law of thermodynamics with varying particle number

$$\mathrm{d}U = T\mathrm{d}S - p\,\mathrm{d}V + \mu\,\mathrm{d}N$$

The grand potential

$$\Omega = F - \mu N = U - TS - \mu N\,, \qquad \mathrm{d}\Omega = -S\mathrm{d}T - p\,\mathrm{d}V - N\mathrm{d}\mu$$

The one-particle statistical weight and density of modes in d-dimensions

$$\mathrm{d}\Gamma = \frac{\mathrm{d}^d r\,\mathrm{d}^d p}{\mathrm{h}^d} \quad \text{and} \quad n_k^d = \left(\frac{L}{2\pi}\right)^d$$

Relation of canonical partition function to thermodynamic functions

$$F(T,V,N) = -\mathrm{k_B} T \ln Z(T,V,N) \qquad U(T,V,N) = \frac{\partial}{\partial\beta}(\beta F)$$

Gamma function and Stirling's approximation

$$\Gamma(N+1) = N! = \int_0^\infty \mathrm{d}x\, x^N\, \mathrm{e}^{-x} \quad \text{and} \quad N! \approx \left(\frac{N}{\mathrm{e}}\right)^N \Leftrightarrow \ \ln N! \approx N(\ln N - 1)\,.$$

Gaussian integral

$$\int_{-\infty}^{\infty} \mathrm{e}^{-\alpha x^2}\mathrm{d}x = \sqrt{\frac{\pi}{\alpha}}\,.$$

Physical Constants and Units

Acceleration due to gravity	g	$9.81\,\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
Ice point	T_{ice}	$273.15\,\mathrm{K}$
Avogadro constant	N_A	$6.022 \times 10^{23}\,\mathrm{mol^{-1}}$
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	$8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$
Stefan constant	σ	$5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\,\mathrm{m^{-1}}$
	$R_\infty hc$	$13.606\,\mathrm{eV}$
Planck constant	h	$6.626 \times 10^{-34}\,\mathrm{J\,s} \equiv 4.136 \times 10^{-15}\,\mathrm{eV\,s}$
$h/2\pi$	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s} \equiv 6.582 \times 10^{-16}\,\mathrm{eV\,s}$
Speed of light *in vacuo*	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
	$\hbar c$	$197.3\,\mathrm{MeV\,fm}$
Charge of proton	e	$1.602 \times 10^{-19}\,\mathrm{C}$
Mass of electron	m_e	$9.109 \times 10^{-31}\,\mathrm{kg}$
Rest energy of electron		$0.511\,\mathrm{MeV}$
Mass of proton	m_p	$1.673 \times 10^{-27}\,\mathrm{kg}$
Rest energy of proton		$938.3\,\mathrm{MeV}$
One atomic mass unit	u	$1.66 \times 10^{-27}\,\mathrm{kg}$
Atomic mass unit energy equivalent		$931.5\,\mathrm{MeV}$
Electric constant	ϵ_0	$8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
Bohr magneton	μ_B	$9.274 \times 10^{-24}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Nuclear magneton	μ_N	$5.051 \times 10^{-27}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	7.297×10^{-3}= 1/137.0
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12}\,\mathrm{m}$
Bohr radius	a_0	$5.2918 \times 10^{-11}\,\mathrm{m}$
angstrom	Å	$10^{-10}\,\mathrm{m}$
barn	b	$10^{-28}\,\mathrm{m^2}$
torr (mm Hg at $0\,^\circ$C)	torr	$133.32\,\mathrm{Pa}$ ($\mathrm{N\,m^{-2}}$)

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Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

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